MMWAVE RadarSS Release Notes

1 Introduction

RadarSS firmware is responsible for configuring RF/analog and digital front-end in real-time. It also schedules periodic temperature based calibrations and monitors based on use need. This enables the mm-Wave front-end to be autonomous and capable of adapting itself to handle temperature and ageing effects, and to enable significant ease-of-use.

2 Release Overview

2.1 Platform and Device Support

The device and platforms supported with this release include:

Supported Devices	Release Status	Supported EVMs
xWR294x ES2.0	Production Release	xWR294x EVM

NOTE: Please refer mmwave_radarss_xWR2x4xP_release_notes.pdf for xWR2x4xLC devices

2.2 Release version

Version	Type
ROM: 2.4.5.3 **PATCH: 2.4.11.0**	Binary patch (9.94KB)

3 Release Contents

3.1 xWR294x ES1.0 RAM contents

3.1.1 DFP 2.4.0.1 Features

- xWR294x is TIs 77GHz RF CMOS Radar SOC. Features supported in this firmware release:
    - This release is derived from AWR2243 DFP2.2 release baseline and backward compatible to DFP 2.2 supported APIs with minor updates for enhancement.
    - Supports 4 TX and 4 Real-only RX chains.
    - Synthesizer RF frequency supported 76 – 81GHz
        - VCO1: 76 – 77GHz
        - VCO2: 76 – 81GHz
    - Supports 7 functional profiles.
    - Supports 266MHz/us max slope
    - Supports programmable filter (not characterized in this engineering release)
- Calibrations
    - APLL calibration.
    - SYNTH calibration.
    - Other calibrations are **NOT** supported in this wakeup release.
- Monitoring
    - Digital / Analog-RF monitors are **NOT** supported in this wakeup release.
- Refer mmWave-Radar-Interface-Control.pdf (ICD) for more information on xWR294x ES1.0 API details.

3.1.2 DFP 2.4.0.2 Features and Enhancements.

- Support for other boot and run-time calibrations
    - LO-DIST calibration.
    - ADC-DC calibration.
    - RX HPF calibration.
    - RX LPF calibration.
    - PD calibration.
    - TX power calibration.
    - RX gain calibration.

    - TX phase shifter calibration.
- Support for calibration data save/restore feature.

3.1.3 DFP 2.4.1.1 Features and Enhancements.

- Support for Digital boot and periodic monitors.
- Support for Analog/RF periodic monitors.
- Support for runtime TX phase shifter calibration.
- Support for new VMON configuration API.
- Improved IFA bias settings for low power ADC modes.
- New Bus safety and ECC aggregator boot monitors.

3.1.4 DFP 2.4.2.1 Features and Enhancements.

- Support for advanced chirp configuration APIs.
- MPU digital boot monitor added.
- New API to add override and dither configurations for RF monitors.
- Improvements in boot time TX power calibration.
- Improvements in RX gain calibration.
- New and optimized TX PS DAC monitor.
- New inter-burst power save options (APLL, LDO duty cycling).
- Improvements in the RF PVT LUTs for reducing power consumption.
- Optimizations in TX monitors for reducing power consumption.

3.1.5 DFP 2.4.2.2 Features and Enhancements.

- No new features/enhancements when compared to DFP 2.4.2.1

3.1.6 DFP 2.4.3.1 Features and Enhancements.

- Support for dynamic per-chirp phase shifter configuration API for 4TX.
- Support for enabling the RSS logger through an API field.
- Improved SYNTH performance by VCO specific configurations.
- Improvements to the TX power calibration accuracy.
- Support for 3 new loopback frequencies in RF loopback.
- Improved TX output power performance at the higher end of the RF BW (81GHz).
- Support for API based control of the SYNTH fast reset end time within the chirp idle time.

- Accuracy improvements in boot and periodic TX phase shifter calibration.
- Blocking the below unsupported periodic analog monitors in AWR_MONITOR_ANALOG_ENABLES_CONF_SB API.
    - RX noise monitor.
    - External analog signals monitor.
    - RX mixer power monitor.
    - RX signal and image band monitor.

3.1.7 DFP 2.4.5.0 Features and Enhancements.

- Improvements to SYNTH frequency error monitoring.
- Improved APLL/SYNTH BW settings.
- BSS logger data can now be sent to a configurable MSS memory buffer from which it can be transmitted out through peripherals like UART/Ethernet.
- Support for digital TX frequency shifting even when loopback is enabled.
- New fault injection sequences for supply LDO and VCO open loop faults.
- Improved error response to VMON self-test failures.
- Improved TX PS boot time calibration.
- New TX loopback sequence updates for ES2.0.
- New RX IF settings for ES2.0.
- Temperature based VCO LDO output voltage settings.

3.2 xWR294x ES2.0 Patch contents

xWR294x ES2.0 has DFP 2.4.5.0 contents in ROM and the below DFP releases are applicable only for ES2.0 devices through firmware patch updates.

3.2.1 DFP 2.4.6.0 Features and Enhancements.

- Improved temperature sensor accuracy by using an EFUSE based two point trimming process.
- Improved RF loopback isolation leading to better calibration and monitoring performance in ES2.0 devices.
- Improved inter-RX and inter-TX mismatches on ES2.0 devices.
- Improved VCO LDO settings for ES2.0 devices.
- Improved RX gain accuracy for ES2.0 devices.
- Improved TX gamma measurement accuracy for ES2.0 devices.

3.2.2 DFP 2.4.8.1 Features and Enhancements.

- Improved RX RF gain targets for better RX gain accuracy on ES2.0 devices.
- Improved TX power monitoring stability.

3.2.3 DFP 2.4.9.1 Features and Enhancements.

- Improved timings for periodic TX phase shifter calibration.
- Improved temperature sensor accuracy.

3.2.4 DFP 2.4.14.0 Features and Enhancements.

- Support for multiple profiles in synth frequency monitor (Live Mode).
- Improved minimum chirp cycle time for advanced chirp configuration by allowing enable/disable control for parameters in advanced chirp configuration.

3.2.5 DFP 2.4.17.0 Features and Enhancements

- No changes to patch binary from DFP 2.4.14.0
- Refer 3.3.12 in Feature/Change List and 4.1.1 in Unsupported features and APIs for documentation updates

3.2.6 DFP 2.4.18.0 Features and Enhancements

- Fixes for Group1 ECC Aggregator SEC and DED error

3.2.7 DFP 2.4.18.1 Features and Enhancements

- No changes to patch binary from DFP 2.4.18.0
- Refer 3.3.14 in Feature/Change List, 4.1.1 in Unsupported features and APIs and 5 Known Issues for documentation updates

3.3 Feature/Changes List

3.3.1 Changes in DFP 2.4.0.1 release (with respect to DFP 2.2.2.1 / AWR2243)

Item type	Key	Description
Feature	MMWAVE_DFP-969	xWR294x BSS firmware migration from AWR2243 DFP 2.2.
Feature	MMWAVE_DFP-971	xWR294x mailbox driver implementation.
Feature	MMWAVE_DFP-1046	Updated RSS interrupts for xWR294x.
Feature	MMWAVE_DFP-1047	APLL calibration in xWR294x.

Feature	MMWAVE_DFP-1048	XTAL detection in xWR294x.
Feature	MMWAVE_DFP-1125	SYNTH calibration in xWR294x.
Feature	MMWAVE_DFP-1049 MMWAVE_DFP-1164	EFUSE based digital and analog trimming in xWR294x.
Feature	MMWAVE_DFP-1051 MMWAVE_DFP-1061 MMWAVE_DFP-1096	Boot time analog initializations in xWR294x like LDO, BFR and other miscellaneous settings.
Feature	MMWAVE_DFP-1052	Updated DCC drivers for the new DCC module in xWR294x.
Feature	MMWAVE_DFP-1054 MMWAVE_DFP-1058 MMWAVE_DFP-1059 MMWAVE_DFP-1106	DFE, FRC, ADCBUF, RAMPGEN, SOCC and other digital initializations in xWR294x.
Feature	MMWAVE_DFP-1070	Support for TX4 chain in xWR294x.
Feature	MMWAVE_DFP-1102	ESM updates for xWR294x.
Feature	MMWAVE_DFP-1119	Clock configurations for xWR294x.
Feature	MMWAVE_DFP-1163	Firmware design update to accommodate up to 7 functional profiles.
Feature	MMWAVE_DFP-1165 MMWAVE_DFP-1166 MMWAVE_DFP-1167	RX-TX channel configuration related API updates for xWR294x.
Feature	MMWAVE_DFP-1169	Temperature sensor updates in xWR294x.
Feature	MMWAVE_DFP-1170 MMWAVE_DFP-1171 MMWAVE_DFP-1172 MMWAVE_DFP-1175	Profile, continuous streaming and chirp configuration API updates in xWR294x.
Feature	MMWAVE_DFP-1173 MMWAVE_DFP-1174	RX and TX gain temperature LUT API updates in xWR294x.
Feature	MMWAVE_DFP-1181 MMWAVE_DFP-1182 MMWAVE_DFP-1183	RF and IF loopback sequences updated for xWR294x.
Feature	MMWAVE_DFP-1541	Defeaturing the RF and PA LDO bypass options in xWR294x.
Feature	MMWAVE_DFP-1545	Updated 20G SYNC LUT settings for xWR294x.
Bug	MMWAVE_DFP-1159	RSS mailbox memory initialization was not performed before use. This has been fixed now.

3.3.2 Changes in DFP 2.4.0.2 release (with respect to DFP 2.4.0.1)

Item type	Key	Description
Feature	MMWAVE_DFP-1123	Support for HPF fast init.
Feature	MMWAVE_DFP-1149	DC BIST updates for AWR29xx.
Feature	MMWAVE_DFP-1176	TX power calibration.
Feature	MMWAVE_DFP-1177	LODIST calibration.
Feature	MMWAVE_DFP-1178	RX gain calibration.

Feature	MMWAVE_DFP-1179	HPF calibration.
Feature	MMWAVE_DFP-1180	LPF calibration.
Feature	MMWAVE_DFP-1184	Multi-TX enable configuration in TX power calibration.
Feature	MMWAVE_DFP-1185	PD calibration.
Feature	MMWAVE_DFP-1186	ADC-DC calibration.
Feature	MMWAVE_DFP-1187	TX phase shifter calibration.
Feature	MMWAVE_DFP-1191	Calibration data save/restore feature support.
Feature	MMWAVE_DFP-1192	Low Power ADC mode support.
Feature	MMWAVE_DFP-1218	Support for DFE stats API in AWR29xx.
Feature	MMWAVE_DFP-1224	Firmware design support for more than 32 monitors in AWR29xx.
Bug	MMWAVE_DFP-1576	APLL control voltage limits for edge temperature bins updated.

3.3.3 Changes in DFP 2.4.1.1 release (with respect to DFP 2.4.0.2)

Item type	Key	Description
Feature	MMWAVE_DFP-1050	STC and PBIST updates for xWR294x.
Feature	MMWAVE_DFP-1086	Periodic register readback monitoring.
Feature	MMWAVE_DFP-1130	AWR29xx: VMON implementation
Feature	MMWAVE_DFP-1189	AWR29xx - Runtime Phase-shifter calibration.
Feature	MMWAVE_DFP-1198	AWR29xx - New BSS VMON control API
Feature	MMWAVE_DFP-1200	AWR29xx - ECC aggregator monitor
Feature	MMWAVE_DFP-1201	AWR29xx - Bus safety monitor
Feature	MMWAVE_DFP-1202	AWR29xx - Synth linearity monitor
Feature	MMWAVE_DFP-1203	AWR29xx - Synth frequency monitor (non-live)
Feature	MMWAVE_DFP-1204	AWR29xx - Synth frequency monitor
Feature	MMWAVE_DFP-1205	AWR29xx - DCC monitor
Feature	MMWAVE_DFP-1206	AWR29xx - Temperature Monitor
Feature	MMWAVE_DFP-1207	AWR29xx - RX IF Stage Monitor
Feature	MMWAVE_DFP-1210	AWR29xx - TX Ball Break Monitor
Feature	MMWAVE_DFP-1211	AWR29xx - TX Gain Phase Monitor
Feature	MMWAVE_DFP-1212	AWR29xx - TX Phase Shifter Monitor
Feature	MMWAVE_DFP-1214	AWR29xx - Internal analog signals monitor
Feature	MMWAVE_DFP-1215	AWR29xx - TX Internal Signal Monitor
Feature	MMWAVE_DFP-1537	Updated Tx Power Monitor for xWR294x
Feature	MMWAVE_DFP-1543	Updated fault injection routines for xWR294x.

Bug	MMWAVE_DFP-1587	Fix for intermittent failures seen in boot time TX power calibration.
Bug	MMWAVE_DFP-1624	Fix for RX gain calibration failures at low temperatures.

3.3.4 Changes in DFP 2.4.2.1 release (with respect to DFP 2.4.1.1)

Item type	Key	Description
Feature	MMWAVE_DFP-1219	Support for advanced chirp configuration APIs.
Feature	MMWAVE_DFP-1612	Support for MPU digital boot monitor.
Feature	MMWAVE_DFP-1630	New API for override and dither configuration of RF monitors.
Feature	MMWAVE_DFP-1631	Improvements in boot time TX power calibration.
Feature	MMWAVE_DFP-1657	Improvements in RX gain calibrations.
Feature	MMWAVE_DFP-1643	New and optimized TX PS DAC monitor.
Feature	MMWAVE_DFP-1644 MMWAVE_DFP-1646	Support for inter-burst APLL and LDO duty cycling to reduce power consumption.
Feature	MMWAVE_DFP-1645	Improved RF PVT LUTs to reduce power consumption.
Feature	MMWAVE_DFP-1655	Optimization of TX monitoring sequences to reduce power consumption.
Bug	MMWAVE_DFP-1559	Fix for DFE parity boot monitoring failures.
Bug	MMWAVE_DFP-1628	Fix for APLL control voltage monitoring failures.
Bug	MMWAVE_DFP-1649	Fix for high TX PS monitoring noise and amplitude values.
Bug	MMWAVE_DFP-1662	Fix for DFE statistics report issue.
Bug	MMWAVE_DFP-1663	Fix for large error seen in the first chirp data in the synth frequency linearity monitor results.
Bug	MMWAVE_DFP-1667	Fix for LODIST closed loop calibration seen on weak devices.
Bug	MMWAVE_DFP-1669	Fix for TX PS DAC fault injection issue.

3.3.5 Changes in DFP 2.4.2.2 release (with respect to DFP 2.4.2.1)

Item type	Key	Description
Bug	MMWAVE_DFP-1677	Fix for Ramp-gen lockstep error and firmware corruption in runtime synth frequency monitor.

3.3.6 Changes in DFP 2.4.3.1 release (with respect to DFP 2.4.2.2)

Item type	Key	Description
Feature	MMWAVE_DFP-1682	Support for dynamic per-chirp phase shifter configuration for all 4TX added as a new API (older API for 3TX has been de-featured).

Feature	MMWAVE_DFP-1684	Support for enabling the RSS logger through an API field. The RSS logger is now disabled by default.
Enhancement	MMWAVE_DFP-1690	Improved SYNTH performance by VCO specific RTRIM and ICP trim configurations.
Enhancement	MMWAVE_DFP-1631	Improvements to the TX power calibration accuracy.
Enhancement	MMWAVE_DFP-1678	Support for 3 new frequencies for the RF loopback APIs.
Enhancement	MMWAVE_DFP-1681	Improved TX output power performance at the higher end of the RF BW (81GHz).
Feature	MMWAVE_DFP-1697	Support for API based control of the SYNTH fast reset end time within the chirp idle time.
Enhancement	MMWAVE_DFP-1710	Blocking unsupported periodic analog monitors in AWR_MONITOR_ANALOG_ENABLES_CONF_SB API. (RX noise monitor, external analog signals monitor, RX mixer power monitor and RX signal and image band monitors)
Bug	MMWAVE_DFP-1638	Fix for intermittent failures seen in periodic SYNTH calibration.
Bug	MMWAVE_DFP-1601 MMWAVE_DFP-1701	RX gain calibration performance has been improved from previous DFP release by corrections to RX gain estimation and IFA gain configuration functions in the firmware.
Bug	MMWAVE_DFP-1625	Improvements to the SYNTH frequency error monitoring.
Bug	MMWAVE_DFP-1698	Improvements to TX phase mismatch between channels.
Bug	MMWAVE_DFP-1674	TX phase shifter accuracy has been improved from previous DFP release but their performance is still under characterization and validation.
Bug	MMWAVE_DFP-1702	RX phase mismatches have been improved from previous DFP release by corrections to RX LO distribution bias adjustment functions in firmware.
Bug	MMWAVE_DFP-1704	When the profile center frequency is less than 200MHz away from the configured cal-mon frequency upper limit, the TX CLPC wasn't executed and only OLPC was performed. This resulted in impaired performance at the top end of the RF BW (81GHz). This scenario is now fixed by limiting the calibration frequency to (cal-mon frequency upper limit – 200MHz) and executing the TX CLPC instead of skipping it.
Bug	MMWAVE_DFP-1696	Fix for the issue where the RSS, during its boot, could overwrite register configurations done by the MSS application to bring the external analog signals to the MSS GPADC.

3.3.7 Changes in DFP 2.4.5.0 release (with respect to DFP 2.4.3.1)

Item type	Key	Description
Enhancement	MMWAVE_DFP-1671	Improved APLL and SYNTH BW settings.
Feature	MMWAVE_DFP-1223	MSS memory buffer based BSS logger data transfer feature.
Enhancement	MMWAVE_DFP-1717	Support for configurable TX digital frequency shift even when loopback is enabled.
Enhancement	MMWAVE_DFP-1708	New sequence for SUPPLY LDO fault injection.

Enhancement	MMWAVE_DFP-1707	New sequence for VCO open loop fault injection.
Enhancement	MMWAVE_DFP-1744	Improved error response to VMON self-test failures.
Enhancement	MMWAVE_DFP-1756	Improved TX PS boot calibration flow.
Enhancement	MMWAVE_DFP-1768	Updated TX loopback sequence for ES2.0.
Enhancement	MMWAVE_DFP-1771	Updated RX IF settings for ES2.0.
Enhancement	MMWAVE_DFP-1773	Temperature based VCO LDO VOUT setting for better synthesizer performance.
Bug	MMWAVE_DFP-1709	Fixed an issue where the TX3 digital frequency shift gets overwritten by the TX4 configuration.
Bug	MMWAVE_DFP-1725	Fix for marginal failures seen in VCO1 control voltage monitoring.
Bug	MMWAVE_DFP-1665	Fix for an error in the sequencer override settings for RX RF and BB in TX loopback-based monitors.
Bug	MMWAVE_DFP-1611	Minor updates to boot time PD calibration and RX gain phase monitor timings.
Bug	MMWAVE_DFP-1742	Fix for the issue where using a Forced Run Cal temperature index of ‘3’ for TX phase shifter calibration failed.
Bug	MMWAVE_DFP-1695	Fix for the issue where the BFR settings in RADARSS overwrote some of analog mux registers that were being used in the MSS GPADC.
Bug	MMWAVE_DFP-1751	Fix for incorrect saturation value in TX OLPC LUTs.
Bug	MMWAVE_DFP-1752	Fixes for static analysis violations.
Bug	MMWAVE_DFP-1758	Increased HPF fast init pulse width to 2.5uS.
Bug	MMWAVE_DFP-1767	Fixed sequence for RAMPGEN lockstep self-test failures.
Bug	MMWAVE_DFP-1770	Fixed default BFR settings for the APLL ICP.
Bug	MMWAVE_DFP-1705	Fixed the issue where the RF1/RF2 VMON controls and status fields were swapped.

3.3.8 *Changes in DFP 2.4.6.0 release (with respect to DFP 2.4.5.0)*

Item type	Key	Description
Bug	MMWAVE_DFP-1784	Fix for RX gain and noise power / loopback power failures.
Bug	MMWAVE_DFP-1783	Fix for the first chirp ADC data corruption on rare devices due to ADC DIG LDO boost settings.
Bug	MMWAVE_DFP-1776	Fixed the spur observed when using SYNTH live monitoring by dithering the RESET_SOCC_END_TIME.
Bug	MMWAVE_DFP-1796	Extended support for 7 functional profiles in the programmable filter configuration API.
Bug	MMWAVE_DFP-1782 MMWAVE_DFP-1831	Zeroed out TX/RX mismatch correction factors in ES2.0.
Bug	MMWAVE_DFP-1800	Improved temperature sensor accuracy when using 0-pt trim default values in firmware.
Bug	MMWAVE_DFP-1801	Updates to HPF fast init start time when using dynamic inter-chirp power save configurations.
Bug	MMWAVE_DFP-1781 MMWAVE_DFP-1838	VCO2 range is now reduced from 5GHz to 4.5GHz. The range can now be chosen between 76-80.5GHz or 76.5-81GHz using a configuration bit in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API. Please refer to the Interface control document for more

		information.
Bug	MMWAVE_DFP-1822	TX internal signals monitor, RX internal signals monitor and PMCLKLO internal signals monitor were using incorrect monitoring frequency. This has now been fixed.
Bug	MMWAVE_DFP-1798	The programmed RAMPEND time is now reduced by 1LSB before programming into the hardware to workaround the hardware issue of counting 1LSB extra during the ramp.
Enhancement	MMWAVE_DFP-1819	Enabled ES2 specific analog improvements related to RX IFA and RF loopback isolation.
Enhancement	MMWAVE_DFP-1833	Improved VCO LDO settings for ES2.0 devices.
Enhancement	MMWAVE_DFP-1845	Improved RX gain accuracy by updating the correct QPSK PD to LNA path loss for ES2.0 devices.
Enhancement	MMWAVE_DFP-1843	Improved TX power reported gamma accuracy for ES2.0 devices.
Enhancement	MMWAVE_DFP-1844	Enabled debug logging of the open loop APLL control voltage during boot calibration.

3.3.9 Changes in DFP 2.4.8.1 release (with respect to DFP 2.4.6.0)

Item type	Key	Description
Enhancement	MMWAVE_DFP-1850	Improved RX RF gain targets for ES2.0 devices
Enhancement	MMWAVE_DFP-1924	Updated the RX package loss to be more accurate on ETS devices.
Bug	MMWAVE_DFP-1785	Fixed stability issues in TX power measurements during calibration and monitoring sequences.
Bug	MMWAVE_DFP-1858	Fixed an issue where the chirp index for phase shifter setting is incorrectly set when using non-zero chirp start index in a burst.
Bug	MMWAVE_DFP-1872	Fixed an issue where the periodic config register read monitor fails when RSS dynamic frequency switching feature is enabled.
Bug	MMWAVE_DFP-1889	MISRA and static analysis violations fixed.
Bug	MMWAVE_DFP-1931	GPADC 10M clock related spur mitigated by keeping the clock disabled when not in use.

3.3.10 Changes in DFP 2.4.9.1 release (with respect to DFP 2.4.8.1)

Item type	Key	Description
Bug	MMWAVE_DFP-1923 MMWAVE_DFP-1951	Fix for the intermittent issue of missing mailbox interrupt that can occur when two or more sub-systems are communicating simultaneously with RSS. The associated retry failure has also been fixed.
Bug	MMWAVE_DFP-1947	Fix for VCO buffer swing marginality seen on weak devices at the higher end of the VCO2 range (~81GHz).
Bug	MMWAVE_DFP-1948	DFE memory PBIST boot monitor was incorrectly executing PBIST on the MSS GPADC memory instead of the BSS GPADC memory. This is now fixed.
Bug	MMWAVE_DFP-1972	Dynamic switching of the GPADC clock introduced inaccuracies in GPADC reads done as part of calibrations and monitoring. This has been fixed by keeping the clock continuously on during calibrations and monitoring. It is turned

		off only during the burst of functional chirps and immediately turned back on.
Enhancement	MMWAVE_DFP-1944	Improved the execution time of periodic runtime phase-shifter calibration by removing redundant computations. No performance changes are expected from this update.
Enhancement	MMWAVE_DFP-1961	Improved temperature sensor accuracy by adding correction factors for the EFUSE hot temperature trim values.

3.3.11 Changes in DFP 2.4.14.0 release (with respect to DFP 2.4.9.1)

Item type	Key	Description
Bug	MMWAVE_DFP-2019	Fix for SYNTH_VCO_OPENLOOP fault injection failure at cold temperatures.
Bug	MMWAVE_DFP-2013	Fix for LODIST calibration failure at hot temperatures
Enhancement	MMWAVE_DFP-1936	Synthesizer frequency monitor (Live Mode) support for multiple profiles
Enhancement	MMWAVE_DFP-1846	Improved minimum chirp cycle time for advanced chirp configuration by allowing enable/disable control for parameters in advanced chirp configuration
Bug	MMWAVE_DFP-2062	Fix for synthesizer frequency monitor failure at cold temperatures.
Bug	MMWAVE_DFP-2068	Fix for Synth VCO LDO short circuit fault at hot temperatures

3.3.12 Changes in DFP 2.4.17.0 release (with respect to DFP 2.4.14.0)

Item type	Key	Description
Bug	MMWAVE_DFP-2144	OSCLKOUT is not supported in xWR294x

3.3.13 Changes in DFP 2.4.18.0 release (with respect to DFP 2.4.17.0)

Item type	Key	Description
Bug	MMWAVE_DFP-2169	Fix to generate ECC_AGC_DED_ERROR when a single bit error occurs in BSS Static RAM. Device now enters safe state when ECC_AGG_SEC_ERROR occurs due to a single bit

		error in BSS Static RAM or a double bit error ECC_AGG_DED_ERROR occurs in BSS Mailbox / BSS Static RAM. Refer device Errata for more details
Bug	MMWAVE_DFP-2290	Fix to clear ESM Group1 ECC aggregator SEC error. When the ESM Group 1 error ECC_AGG_SEC_ERROR occurs for the first time, single bit error is corrected and async event AWR_AE_RF_ADV_ESMFAULT_STATUS_SB is generated. Subsequent single bit error event would correct the single bit error but async event would not get generated. This has been fixed now

3.3.14 Changes in DFP 2.4.18.1 release (with respect to DFP 2.4.18.0)

Item type	Key	Description
Bug	MMWAVE_DFP-2395	Defeature “Rx FE disabled”, value 4 in LOOPBACK_SEL field of AWR_LOOPBACK_BURST_CONF_SET_SB. Added 20MHz IFA loopback frequency in IF_LOOPBACK_FREQ field of AWR_LOOPBACK_BURST_CONF_SET_SB.

4 Unsupported features and APIs (applicable to all DFP releases)

The following APIs and features are not validated fully at system level, it is recommended not to use these APIs in this and all previous DFP releases. This list of unsupported features is in addition to the list mentioned in known issues.

4.1 Unsupported Features

4.1.1 Functional APIs

API	Feature	Description
PCR self-test feature in AWR MONITOR_RF_DIG_LATENTFAULT_CONF_SB	PCR self-test	The PCR self-test is not supported in xWR294x.
AWR_INTERCHIRP_BLOCKCONTROLS_SB	Inter-chirp power saving timing configurations	This API is not validated at system level. It is recommended not to use the same.
AWR_MONITOR_RX_NOISE_FIGURE_CONF_SB	RX noise figure monitor	The RX noise figure monitor is not supported in xWR294x.
AWR_MONITOR_RX_MIXER_IN_POWER_REPORT_AE_SB	Rx mixer input power monitor	The RX mixer input power monitor is not supported in xWR294x.
TXn POWER_BACKOFF fields in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB	Calibration and Monitoring Frequency and TX Power limits	Recommended to perform factory calibration with 0dB back-off set in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API and, store and restore

	API	the calibration data from non-volatile memory. With this option, user can set nonzero back-off option in AWR_CAL_MON_FREQUENCY_TX_POWER_LIMITS_SB API during normal runs to back-off monitor/calibration chirp's power.
AWR_MONITOR_VMON_CONF_SB	VMON configuration	This feature has not been validated at a system level. It is not recommended to use this.
SYNTH_VCO1_SLOPE and SYNTH_VCO2_SLOPE fields in the PLL_CONTROL_ VOLTAGE_VALUES in AWR_MONITOR_PLL_CONTROL_VOLTAGE_REPORT_AE_SB	VCOx slope monitoring	This feature is not supported in xWR294x.
LPF_CUTOFF_BANDEDGE_DROOP_VALUE_RX0, LPF_CUTOFF_STOPBAND_ATTEN_VALUE, LPF_CUTOFF_BANDEDGE_DROOP_VALUE_RX fields in **AWR_MONITOR_RX_IFSTAGE_REPORT_AE_SB** ASYCNC report	LPF monitoring results ASYNC report	These fields are only supported for debug purposes and should not be used for production use-cases.
LPF_CUTOFF_BANDEDGE_DROOP_THRESH and LPF_CUTOFF_STOPBAND_ATTEN_THRESH fields in **AWR_MONITOR_RX_IFSTAGE_CONF_SB** API	LPF monitoring threshold fields in the IF STAGE monitoring configuration API	These fields are only supported for debug purposes and should not be used for production use-cases.
Some combinations of RX_GAIN and RF_GAIN_TARGET in PF_RX_GAIN field of the **AWR_PROFILE_CONF_SB** and **AWR_CONT_STREAMING_MODE_CONF_SET_SB** APIs.	(RX Gain, RF Gain Target) (44dB, 32dB)	Since the maximum IF gain is +10dB in xWR294x, the specified pairs of (RX Gain, RF Gain Target) settings are not supported.
Some values of the GLOBAL_RESET_MODE field of the **AWR_ADVANCE_CHIRP_CONF_SB** API are not supported when using Advanced Frame Configuration.	Value 0 - "End of frame"	When using advanced frame configuration in conjunction with advanced chirp configuration, the "end of frame" reset mode is not supported in advanced chirp config API.
OSCCLKOUT_DIS in CASCADING_ PINOUTCFG field of the AWR_CHAN_CONF_SET_SB API.	OSCCLKOUT	OSCCLKOUT is not supported. It is recommended to keep OSCLKOUT disabled.
RX FE disabled mode. Value 4 in LOOPBACK_SEL field of AWR_LOOPBACK_BURST_CONF_SET_SB	RX FE disabled mode in loopback burst configuration API	RX FE disabled option in not supported for loopback burst configuration API

4.1.2 Debug APIs

API	Feature	Description
AWR_RF_PALOOPBACK_CFG_SB AWR_RF_LOLOOPBACK_CFG_SB AWR_RF_IFLOOPBACK_CFG_SB	Loopback enables	PA, LO and IF loopback APIs are not supported in functional mode, recommended to use only for debug.
AWR_RF_TEST_SOURCE_CONFIG_SET_SB AWR_RF_TEST_SOURCE_ENABLE_SET_SB	Test source feature	Test source feature is not supported in functional mode, recommended to use only for debug.
AWR_CONT_STREAMING_MODE_CONF_SET_SB AWR_CONT_STREAMING_MODE_EN_SB	Continuous streaming mode	Continuous streaming mode is not supported in functional mode, recommended to use only for debug.
OSCCLKOUT_DIS in CASCADING_PINOUTCFG field of the AWR_CHAN_CONF_SET_SB API is not supported.	OSCCLKOUT	It is recommended to keep OSCLKOUT disabled.

5 Known Issues

Key	Severity	Description
MMWAVE_DFP-1649	Minor	**Noise metric indicators vary with temperature.** RF loopback-based TX and RX gain phase mismatch monitors include a noise metric indicator to enable detection of corruption by external interference. The noise metric varies with temperature. Strategy to determine thresholds for detecting interference is mentioned in a monitoring application note. The strategy includes temperature dependent thresholds.
MMWAVE_DFP-1560	Minor	**DCC clock monitor failures for RCOSC_10M** The DCC clock monitor for the RCOSC_10M clock source can fail on some of the devices due to variations higher than +/- 30% in the RCOSC frequency and the strict 30% threshold check in FW AWR_MONITOR_DUAL_CLOCK_COMP_REPORT_AE_SB monitoring report. **Workaround:** It is recommended to implement a new threshold check of +/-50% in the application software for RCOSC_10M frequency in case any error in the status (STATUS_CLK_PAIR4) flag in AWR_MONITOR_DUAL_CLOCK_COMP_REPORT_AE_SB monitoring report.
MMWAVE_DFP	Minor	**Minimum RX Gain setting updated from 24dB to 28dB** RX Gain settings 24dB and 26dB show high inaccuracy at cold

-2087		temperatures. **Workaround:** It is recommended to update the RX Gain setting to a minimum of 28dB in the following API's • AWR_PROFILE_CONF_SB • AWR_LOOPBACK_BURST_CONF_SET_SB • AWR_CONT_STREAMING_MODE_CONF_SET_SB
MMWAVE_DFP -2105	Major	**ECC_AGC_DED_ERROR mapped to ESM_GROUP1_ERRORS_LSB** Double bit ECC errors in BSS static RAM and BSS mailbox indicated by ECC_AGC_DED_ERROR is mapped to ESM_GROUP1_ERRORS_LSB. ESM_GROUP1_ERRORS are group of non-fatal errors, however double bit ECC error ECC_AGC_DED_ERROR is a fatal error. **Workaround:** **It is recommended to treat** ECC_AGC_DED_ERROR in field ESM_GROUP1_ERRORS_LSB of AWR_AE_RF_ADV_ESMFAULT_STATUS_SB and AWR_RF_ADV_ESMFAULT_STATUS_SB should be treated as a fatal error
MMWAVE_DFP -2395	Major	RX FE Disabled mode, value 4 in LOOPBACK_SEL field of AWR_LOOPBACK_BURST_CONF_SET_SB when used to measure noise floor show random jump in noise floor **Workaround:** It is recommended to use IF loopback mode with an out of band tone frequency to measure noise floor instead of RX FE Disabled mode